CS 3006
Parallel and Distributed Computing
Lecture 2

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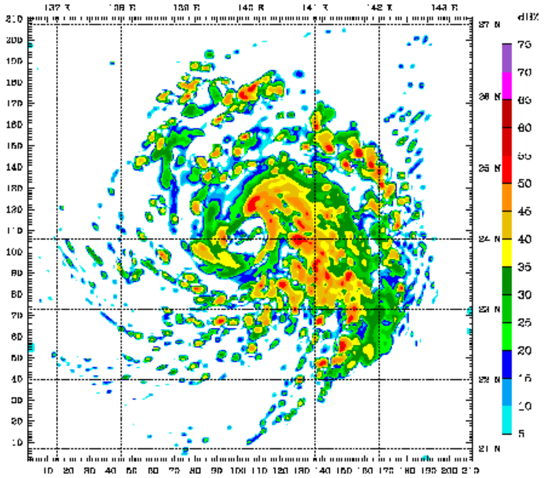
Introduction to Parallel and Distributed Computing

Lecture’s Agenda

- **Motivating Parallelism**
- **Systems vs Computing**
- **Parallel vs Distributed Computing**
- **Practical Applications of Parallel and Distributed Computing**
- **Summary**
- **Additional Resources**

Motivation for Parallel and Distributed Computing

- Uniprocessor are fast but
  - Some problems require too much computation
  - Some problems use too much data
  - Some problems have too many parameters to explore
- For example
  - Weather simulations, online gaming, web servers, code breaking



Motivating Parallelism

- **Developing parallel hardware and software has traditionally been time and effort intensive**
- **If one is to view this in the context of rapidly improving uniprocessor speeds, one is tempted to question the need for parallel computing**
- **Latest trends in hardware design indicate that uniprocessors are not able to sustain the rate of realizable performance increments in the future**
- **A number of fundamental physical and computational limitations**
- **The emergence of standardized parallel programming environments, libraries, and hardware have significantly reduced time to develop (parallel) solution**

Motivating Parallelism (Cont.)

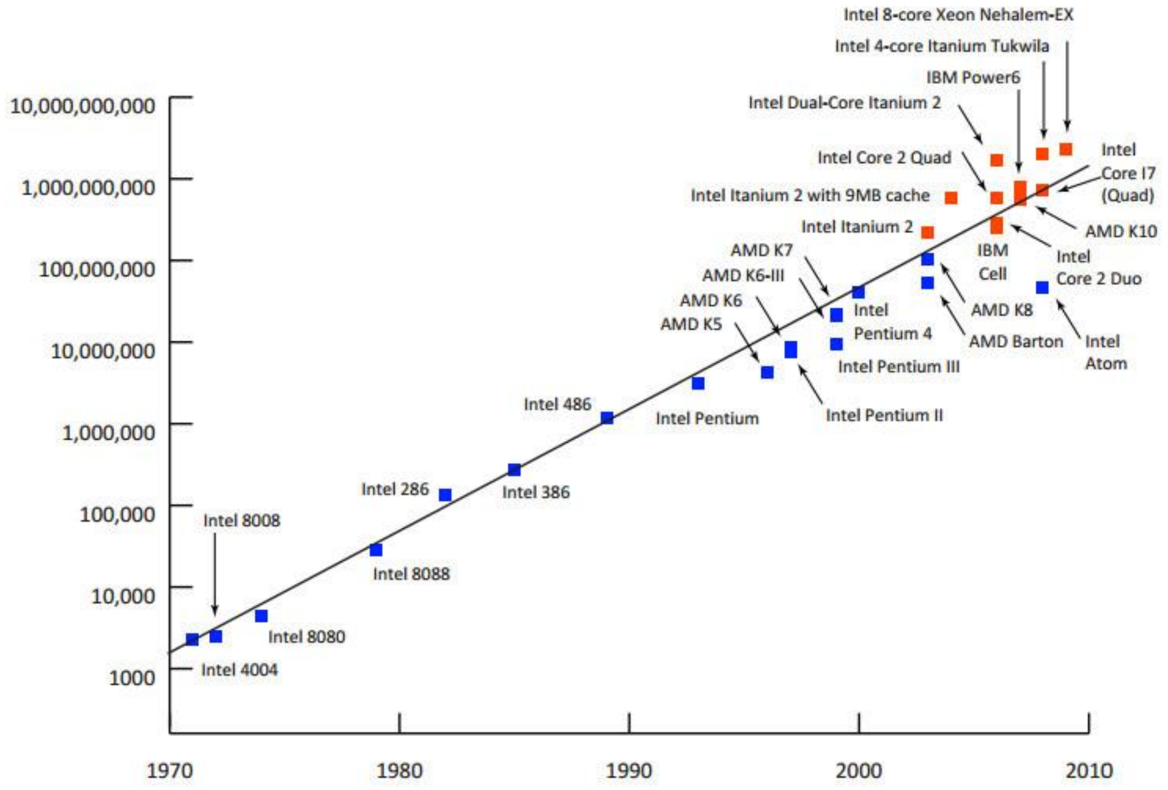
What is Moore's Law?

- **Proposed by Gorden E. Moore in 1965 and revised in 1975**
- **It states that [Simplified Version]**
  - **"Processing speeds, or overall processing power for computers will double every 18 months."**
- **A more technically correct interpretation**
  - **"The number of transistors on an affordable CPU would double every 18 months."**

Motivating Parallelism (Cont.)

What is Moore's Law? (Cont.)

- **Number of transistors incorporated in a chip will approximately double every 18 months.**



Motivating Parallelism (Cont.)

What is Moore's Law? (Cont.)

- More computational power implicitly **means** more transistors
- Then **why need** second interpretation?
- Let's have a look on **empirical data** from 1970 to 2009

  - In 1970's (i.e., from 1970 to 1979), processor speeds ranged from 740 KHz to 8 Mhz. Difference shows that **both the interpretations** are correct

  - From 2000 to 2009, Speeds ranged from 1.3 GHz to 2.8 GHz
    - ✓ Speed difference is too low but, number of integrated transistors ranged from 37.5 million to 904 million
    - ✓ So, **second interpretation** is more accurate.

Motivating Parallelism (Cont.)

Why doubling the transistors does not doubles the speed?

- The answer is increase in number of transistor per processor is due to multi-core CPU's
- In multi-core CPUs, speed will depend on the proper utilization of parallel resources
- To follow Moore's law, companies had to:
  - Introduce Ultra Large Scale Integrations (ULSI)
    - ✓ Ultra large-scale integration (ULSI) is the process of embedding millions of transistors on a single silicon semiconductor microchip
  - And multi-core processing era

Motivating Parallelism (Cont.)

Will Moore's law hold forever?

- **Adding multiple cores on single chip causes heat issues**
- **Increasing the number of cores, may not be able to increase speeds [Due to inter-process interactions]**
- **Transistors would eventually reach the limits of miniaturization at atomic levels**

Motivating Parallelism (Cont.)

- **So, we must look for efficient parallel software solutions to fulfill our future computational needs**

- **As stated earlier, number of cores on a single chip also have physical and computational limitations**

- **Solution:**
  - **Need to find more scalable distributed and hybrid solutions**

Motivating Parallelism (Cont.)

The Memory / Disk Speed Argument

- While clock rates of high-end processors have increased at roughly 40% per year over the past decade, DRAM access times have only improved at the rate of roughly 10% per year over this interval

- This mismatch in speeds causes significant performance bottlenecks

- Parallel platforms provide increased bandwidth to the memory system

Motivating Parallelism (Cont.)

The Memory / Disk Speed Argument (Cont.)

- Parallel platforms also provide **higher aggregate caches**
- Some of the fastest growing applications of parallel computing utilize not their raw computational speed, rather their **ability to pump data** to memory and disk faster
  - CPU is fast but memory isn't, which is a **mismatch** in context of parallel programming
  - Solution: Parallel platform (increased **bandwidth** to memory) and combination or collection of caches

Motivating Parallelism (Cont.)

The Data Communication Argument

- As the network evolves, the vision of the Internet as **one large computing platform** has emerged

- In many applications like databases and data mining problems, the **volume of data** is such that they cannot be moved

- Any **analyses** on this data must be performed over the network using parallel techniques

Systems vs Computing

Distributed Systems:

- **"A collection of autonomous computers, connected through a network and distribution middleware."**
- **Enables computers to coordinate their activities and to share the resources of the system**
- **The system is usually perceived as a single, integrated computing facility**
- **Mostly concerned with the hardware-based accelerations**

Systems vs Computing (Cont.)

Distributed Computing:

- **"A specific use of distributed systems to split a large and complex processing into subparts and execute them in parallel to increase the productivity."**

- **Computing mainly concerned with software-based accelerations (i.e., designing and implementing algorithms)**

Parallel vs Distributed Computing

Parallel Computing:

- The term is usually used for developing concurrent solutions for following two types of the systems:
  - ➢ Multi-core Architecture
  - ➢ Many core architectures (i.e. CPU’s & GPU’s)

Distributed Computing:

- The type of computing mainly concerned with developing algorithms for the distributed cluster systems
- Here distributed means a geographical distance between the computers without any shared-Memory

Parallel vs Distributed Computing (Cont.)

Parallel Computing:

- Shared memory
- Multiprocessors
- Processors share logical address spaces
- Processors share physical memory
- Referred to the study of parallel algorithms

Distributed Computing:

- Distributed memory
- Clusters of workstations
- Processors do not share address space
- There is no sharing of physical memory
- Study of theoretical distributed algorithms or neural networks

Practical Applications of P&D Computing

Scientific Applications:

- **Functional and structural characterization of genes and proteins**
- **Applications in astrophysics have explored the evolution of galaxies, thermonuclear processes, and the analysis of extremely large datasets from telescope**
- **Advances in computational physics and chemistry have explored new materials, understanding of chemical pathways, and more efficient processes**

Practical Applications of P&D Computing (Cont.)

Scientific Applications (Cont.):

- Weather modeling for simulating the track of natural hazards like extreme cyclones (storms), flood prediction
- Large Hydron Collider (LHC) at European Organization for Nuclear Research (CERN) generates petabytes of data for a single collision

Practical Applications of P&D Computing (Cont.)

Commercial Applications:

- **Some of the largest parallel computers power Wall Street!**
- **Data mining analysis for optimizing business and marketing decisions**
- **Large scale servers (mail and web servers) are often implemented using parallel and distributed platforms**
- **Applications such as information retrieval and search are typically powered by large clusters**

Practical Applications of P&D Computing (Cont.)

Computer Systems Applications:

- Network intrusion detection: a large amount of data needs to be analyzed and processed
- Cryptography (the art of writing or solving codes) employs parallel infrastructures and algorithms to solve complex codes – SHA 256
- Graphics processing – Each pixel represents a binary
- Embedded systems increasingly rely on distributed control algorithms - Modern automobiles use embedded systems

Limitations of Parallel Computing

- **Parallel computing requires designing the proper communication and synchronization mechanisms between the processes and the sub-tasks.**

- **Exploring the proper parallelism from a problem is a hectic process.**

- **The program must have low coupling and high cohesion however it is difficult to create such programs.**

- **Parallel computing needs relatively more technical skills to code a parallel program.**

Summary

- **Motivation for Parallel and Distributed Computing**
  - **Apps like weather simulations, online gaming, web servers etc. requires PDC**
- **Motivating Parallelism**
  - **Uniprocessors cannot provide required performance due to theirs computational limitations**
  - **Moore's Law - Number of transistors double every 18 months**
    - ✓ **Doubling the transistors does doubles the speed?**
      - **True for single core chip but not for multi-core CPUs**
    - ✓ **Will Moore's law hold forever?**
      - **Heat issues, synchronization issues, limits of miniaturization of transistors**

Summary (Cont.)

- **Motivating Parallelism**
  - **The Memory / Disk Speed Argument**
    - **Speed mismatch between processor and DRAM can be reduced using parallel platforms and aggregated caches**
  - **The Data Communication Argument**
    - **Analyses on large scale data can be performed over the network using parallel techniques**

Summary (Cont.)

- **Distributed System vs Distributed Computing**
  - **Distributed System - Collection of autonomous computers, connected through a network and distribution middleware**
  - **Distributed Computing - Specific use of distributed systems to split a large problem into subparts and execute them in parallel**

- **Parallel Computing vs Distributed Computing**
  - **Parallel Computing - Developing concurrent solutions**
  - **Distributed Computing - Autonomous computers on different locations connected via network**

Summary (Cont.)

- **Practical Applications of P&D Computing**
  - **Scientific Applications - Characterization of genes and proteins, astrophysics applications, computational physics and chemistry, weather modeling, nuclear physics**
  - **Commercial Applications - Processing of stock exchanges, banking sector, business analytics, web servers, mail servers, search engines**
  - **Computer Systems Applications - Network intrusion detection, cryptography, graphic processing, embedded systems control**
- **Limitations of Parallel Computing**
  - **Parallel computing requires designing proper communication and synchronization mechanisms**
  - **Parallel program must have low coupling and high cohesion**

Additional Resources

- **Introduction to Parallel Computing by Ananth Grama and Anshul Gupta**
  - **Chapter 1: Introduction to Parallel Computing**
    - **1.1 Motivating Parallelism**
      - **1.1.1 The Computational Power Argument – from Transistors to FLOPS**
      - **1.1.2 The Memory / Disk Speed Argument**
      - **1.1.3 The Data Communication Argument**
    - **1.2 Scope of Parallel Computing**

Questions?